

Expense Forecasting API (1.0.0)

Download OpenAPI specification: [Download](#)

API untuk prediksi pengeluaran bulanan menggunakan Ensemble Model (ARIMA + SARIMAX + Deep Learning). Upload CSV untuk training atau gunakan endpoint /predict untuk inferensi langsung.

Utility

Health

Cek status server dan model.

Responses

> 200 Successful Response

GET /health



Response samples

200

Content type
application/json

Copy

```
{  
  "status": "string",  
  "model_loaded": true,  
  "version": "1.0.0"  
}
```

Training

Train

Upload CSV dan latih semua model (ARIMA + Deep Learning).

QUERY PARAMETERS

dl_epochs	integer (DL Epochs) [1 .. 500] Default: 80 Jumlah epoch training DL
dl_batch_size	integer (DL Batch Size) [8 .. 256] Default: 32 Batch size DL
arima_max_p	integer (Arima Max P) [1 .. 5] Default: 3 ARIMA max P
arima_max_q	integer (Arima Max Q) [1 .. 5] Default: 3 ARIMA max Q

REQUEST BODY SCHEMA: multipart/form-data
required

file required	string <application/octet-stream> (File) CSV dataset pengeluaran
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Responses

> **200** Successful Response

> **422** Validation Error

POST /train



Response samples

200

422

Content type

application/json

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```
{
  "success": true,
  "message": "string",
  - "arima_order": [
    null
  ],
  "smape_arima": 0,
  "smape_dl_daily": 0,
  "ensemble_forecast": 0,
  "months_in_data": 0,
  "timestamp": "string"
}
```

Inference

Predict

Prediksi pengeluaran sisa bulan ini.

REQUEST BODY SCHEMA: application/json
required

lag_1 required	number (Lag 1) Pengeluaran hari lalu (Rp)
lag_2 required	number (Lag 2) Pengeluaran 2 hari lalu (Rp)
lag_3 required	number (Lag 3) Pengeluaran 3 hari lalu (Rp)
rolling_mean_7 required	number (Rolling Mean 7) Rata-rata 7 hari terakhir (Rp)
transaction_count required	integer (Transaction Count) >= 0 Jumlah transaksi hari ini
day_of_week >	Day Of Week (integer) or Day Of Week (null) (Day Of Week)
month >	Month (integer) or Month (null) (Month)
is_weekend >	Is Weekend (integer) or Is Weekend (null) (Is Weekend)
mtd_progress >	Mtd Progress (number) or Mtd Progress (null) (Mtd Progress)

Responses

> 200 Successful Response

> 422 Validation Error

POST /predict



Request samples

Payload

Content type

application/json

Copy

```
{
  "lag_1": 0,
  "lag_2": 0,
  "lag_3": 0,
  "rolling_mean_7": 0,
  "transaction_count": 0,
  "day_of_week": 6,
  "month": 1,
  "is_weekend": 1,
  "mtd_progress": 1
}
```

Response samples

200

422

Content type

application/json

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```
{
  "success": true,
  "message": "string",
  - "dl_forecast": {
    "forecast_rupiah": 0,
    "confidence_lower": 0,
    "confidence_upper": 0,
    "trend_pct": 0,
    "trend_direction": "string"
  },
}
```

```
"anima_forecast": 0,  
"ensemble_forecast": 0,  
"days_forecasted": 0,  
"timestamp": "string",  
"rolling_mean_30": 0  
}
```

Analysis

Budget Analysis

Analisis budget vs forecast.

REQUEST BODY SCHEMA: application/json

required

— budget_target	number (Budget Target) > 0
required	Target anggaran bulanan (Rp)

Responses

> **200** Successful Response

> **422** Validation Error

POST

/budget-analysis



Request samples

Payload

Content type

application/json

Copy

```
{  "budget_target": 0}
```

Response samples

200

422

Content type

application/json

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```
null
```

Monthly Summary

Ringkasan pengeluaran bulanan.

QUERY PARAMETERS		
→	n_months	integer (N Months) Default: 6

Responses

- > 200 Successful Response
- > 422 Validation Error

GET

/monthly-summary



Response samples

200

422

Content type

application/json

Copy

null